Safety and Efficacy of Ceftaroline in Neonates With Staphylococcal Late-onset Sepsis: A Case Series Analysis

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Treatment of late-onset neonatal staphylococcal sepsis is sometimes challenging with feared side effects of vancomycin, increasing minimal inhibitory concentrations and questions about catheter management. In case of failure, ceftaroline was administered as a compassionate treatment in 16 infants (gestational age of less than 32 weeks and less than 28 postnatal days), whose first-line treatment failed. We report 11 successes and no severe adverse drug reactions. Larger data are required to confirm these encouraging results.

Key Words: preterm, coagulase-negative *Staphylococci*, ceftaroline, late-onset sepsis, persistent bacteremia

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Late-onset sepsis (LOS) with coagulase-negative *Staphylococci* (CoNS) is often catheter-related, and associated with high morbidity in preterm neonates. The incidence is 4–11 per 1000 catheter days.[1] Vancomycin is the first-line treatment because of its narrow spectrum and activity against CoNS. Unfortunately, vancomycin heteroresistant CoNS strains have been increasingly reported worldwide.[2] Despite improvements through pharmacokinetic model-based adjustment, treatment with higher concentration of vancomycin could become necessary to guarantee good efficiency [targeting Air under the curve/minimal inhibitory concentration (MIC) ratio > 400], which might increase ototoxicity and nephrotoxicity. Also, some cases are affected by treatment failure. Daptomycin, linezolid or adjunction of rifampicin, are second-line therapies, are all with notable adverse drug reactions (ADR).[3]

Ceftaroline, a fifth-generation cephalosporin, is active against methicillin-resistant *Staphylococcus aureus* and CoNS and may be an alternative.[4] It has been approved by the European Medicines Agency in neonates from birth for complicated skin and soft tissue infections and community-acquired pneumonia. It has been used in exceptional circumstances, off-label, in preterm infants, and has demonstrated a good safety profile.[5]

In this study, we aimed to evaluate the efficacy and safety of ceftaroline for the treatment of neonatal LOS with *Staphylococci* especially CoNS in infants born at <32 weeks gestational age (GA) and <28 postnatal days whose first-line treatment failed.

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MATERIAL AND METHODS

Study Design and Patients

This single-center retrospective case series study was conducted in a neonatal intensive care unit at Nantes University Hospital, France. Neonates born at less than 32 weeks GA and less than 28 postnatal days presenting LOS with CoNS or *S. aureus* from January 2018 to May 2022 were eligible if treated with ceftaroline after vancomycin failure. The decision to treat was made at a multidisciplinary infectious diseases meeting. The reasons for rotation to ceftaroline could be persistent bacteremia, defined as positive blood cultures more than 48 hours apart despite effective vancomycin trough serum concentrations, acute kidney injury (AKI), defined according to the Kidney Disease: Improving Global Outcomes' classification or clinical deterioration with bacteremia and ventilator-acquired pneumonia (VAP).[6]

This study was observational and used anonymous data. In accordance with French regulations, parents provided informed consent for medical data collection at admission.

Study Assessments

Efficacy evaluation included assessment of clinical and microbiological responses at the end of treatment (EOT) and 7 days later to identify potential recurrence of infection. Clinical cure was defined as sepsis-related therapeutic de-escalation and resolution of physical symptoms permitting discontinuation of antibacterial therapy. Clinical failure was defined as incomplete resolution of physical symptoms or failure of sepsis-related therapeutic de-escalation. Microbiological cure was defined as the eradication of the initial pathogen in blood cultures without early or late recurrence. Microbiological failure was defined as the persistence of the initial pathogen in blood cultures or its early or late recurrence, whether the bacteremia recurred with the same pathogen before or after 72 hours of EOT.

Clinical Analysis

The following clinical data were collected from medical records and monitoring sheets: sex, GA, postmenstrual age, birth weight, weight at the time of infection, coexisting medical condition, vancomycin dose and duration, ceftaroline dose and duration, ceftaroline plasma concentration when available and catheter management (removal). Available biological parameters were reviewed to identify ADRs.

The sepsis-related therapeutic escalation refers to the specific management of sepsis. We recorded respiratory support (high-frequency oscillatory or conventional ventilation), introduction of insulin, fluid boluses and use of vasopressors.

ADRs that occurred after the start of treatment were assessed until the end of hospitalization. We considered ADRs based on those reported in the summary of product characteristics (SPC) of ceftaroline and signals from the pharmacovigilance database Vigilyze (the World Health Organization global individual case safety report database), Uppsala Monitoring Centre (UMC), Uppsala, Sweden, the Micromedex-DRUGDEX and the Martindale databases (IBM watson Health, Greenwood Village, CO). Severity was assessed according to the general guideline of the National Cancer Institute Common Terminology Criteria for Adverse Events (V5) classification and the imputability according to the Naranjo algorithm.[7]

Biologic Analysis

Bacterial isolates from blood cultures were identified using a Vitek Matrix-Assisted Laser Desorption Ionization-Time of Flight (MALDI-TOF) Mass Spectrometer (BioMérieux, Marcy l'Etoile, France). Ceftaroline MICs were determined using the Etest MIC strips, whereas vancomycin MICs were measured using VITEK 2 method (AST-P631 cards, BioMérieux SA, France) and controlled by broth microdilution method (UMIC, Biocentric, France) for MICs values ≥2 mg/L. MICs interpretation was performed according to 2021 the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines.

A vancomycin serum steady-state trough concentration (C_{min}) was considered effective when it was 10-fold above the MIC (arbitrary cutoff used in the unit as target for vancomycin dosage).

Ceftaroline plasma concentrations were measured 6 hours after the start of at least the fifth injection by high-performance liquid chromatography diode array detector method validated according to the European Medicines Agency criteria.

The duration of bacteremia before the introduction of ceftaroline was defined as the time from the first to the last positive blood culture. Time to sterile culture was the time from the first day of ceftaroline administration to the first negative blood culture.

Statistical Methods

Analyses are regarded as descriptive. We used the number, median (min–max) and range to describe the results. Statistical Package for the Social Science for windows (IBM, Version 26.0, Armonk, New York, USA, 2019) was used for the statistical analysis.

RESULTS

Sixteen infants met the inclusion criteria for this study (Table 1). The median GA, weight at birth and chronological age at sepsis were 25 weeks + 1 day (24 + 0-26 + 6), 725 g (485–910) and 8 days (1–25). Fifteen had bacteremia with CoNS, one with *S. aureus*, lasting a median of 5 days (1–11) before the introduction of ceftaroline. Eight patients (50%) had their catheters changed before end of ceftaroline (2 days before to 3 days after initiation, except for patient 13 whose catheter was changed 8 days before). Overall, they received 8 mg/kg/8 h (3–8.5) of ceftaroline with a 1-hour infusion, for 7.5 days (4–13) and overall, 8 days antibiotic therapy (1–11) from the first negative blood culture. Thirteen patients of 16 (81%) had successful treatment at EOT and 11 of 16 patients (69%) had treatment 7 days later, including the one child with *S. aureus*. Sterilization of blood was systematically performed in less than 3 days when treatment was successful. Patient 5 died of refractory bronchopulmonary dysplasia after palliative care at EOT. Patient 7 experienced bacteremia recurrence on the sixth day of ceftaroline treatment. Patient 12 had persistence of the pathogen. Patients 2 and 8 presented with late recurrence. Three of the last four patients did not initially undergo catheter removal despite persistent bacteremia: one because of hemodynamic instability and two in the hope of sterilization in view of the good initial response. Patient 7 experienced treatment failure despite catheter removal and strain with a ceftaroline MIC of 1 mg/L. The four patients were then successfully treated with vancomycin after catheter removal when this had not been done. All the patients with VAP underwent successful treatment.

Nine patients (56%) had AKI and received a median of 6 (3–8.5) mg/kg/8 h at initiation depending on their renal function. Four of five cases of treatment failure had AKI and ceftaroline doses adjusted to renal function.

Ceftaroline C_{min} were available for five patients and were in the range of 3.6–5.9 µg/mL high-performance liquide chromatography diode array detector method resulting in a free-C_{min}/MIC ratio estimated at 2.2 (1.5–15) considering a half-life of 2 hours and plasma protein binding of 20% (5).

All ADRs attributable to ceftaroline are reported in Table 1 (score of Naranjo ≥ 1). There were three cases of necrotizing enterocolitis, classified according to the modified Bell staging, one of which was considered a possible severe ADR. One patient experienced severe cholestasis, significantly worsening concomitantly with ceftaroline. All patients except patient 5 are alive and 7 months to 4 years of age.

DISCUSSION

In this case series reporting mainly very preterm infants who received compassionate use of ceftaroline for a CoNS LOS (15/16), 11 (68.8%) were treated successfully. The existing literature reports only five cases of ceftaroline use in preterm infants born before 32 weeks GA.[5]

Our usual practice tends to treat the patient without changing the catheter, given the difficulties of its management in this population. Therefore, only patient 13 had a catheter change during the first-line treatment. Even if five patients (31%) were treated successfully without catheter removal, three of five therapeutic failures (all with CoNS) could have been avoided because subsequent catheter removal allowed patients to be treated appropriately. This observation reinforces the need to change the catheter in cases of persistent bacteremia, contrary to our usual practice in the hope vancomycin will be sufficient.

One severe necrotizing enterocolitis was reported, and intestinal events were frequent. Ceftaroline has a broad spectrum and its use may have consequences on the intestinal microbiota. Although cephalosporins have been widely used with good safety results, we cannot exclude that necrotizing enterocolitis cases observed in our cohort were in fact induced by ceftaroline. Moreover, two patients experienced mild hypereosinophilia, a probable ADR, reported in the SPC of ceftaroline as rare. The patient presenting with severe cholestasis, an ADR not described before, also had prolonged parenteral nutrition which makes it only a possible ADR.

In the literature, the target for the treatment of community-acquired *S. aureus* skin infections is to maintain a concentration 35% of the time above MIC. In the intensive care unit, some authors recommend a target of free-C_{min}/MIC ratio of up to 6.[8] Three of our successfully treated patients have C_{min}/CMI below this target but maintained above 1, as suggested by some authors. The few dosages did not allow us to conclude whether lower targets could be generalized. Whether the failure is secondary to under dosage is debatable in patient 7: we hypothesized that ceftaroline trough concentration could not be maintained at a sufficient level for eradication at a dose of 6 mg/kg/8 h.

TABLE 1. Description of Population and Results

Patients*	Gender	GA or PMA (Weeks) and Weight (g), respectively, (a) at Birth (b) at Sepsis	Specific Condition	*Staphylococcus* Species and Vancomycin/Ceftaroline MICs (mg/L)	Sepsis Intensity	PCT (ng/mL)	Maximal C-reactive Protein (mg/L)	Medical Reason for Ceftaroline	Duration (Days): Vancomycin	Duration (Days): Effective Vancomycin†	Duration (Days): Bacteriemia Before Ceftaroline	Duration (Days): Time to Sterile Culture After CFT	Duration (Days): Antibiotics After First Negative Culture	Ceftaroline Posology (per dose in mg/kg) Interval (h) Length (d)	Ceftaroline Concentration (µg/mL)	Catheter Removal for Treatment‡	Outcome Microbiological/Clinical at (a) EOT and (b) 7 Days Later	Adverse Effects Possibly Related to Treatment, Grade (G) Severity According to CTCAE General Guidelines, Imputability According to Naranjo
1	Male	(a) 24+5; 715 (b) 25+6; 745	Diaphragmatic hernia	*Staphylococcus capitis* (1.0/0.25) *Staphylococcus haemolyticus* (1.0/1.0)	HFOV Dopa Insuline FB	NR	<5	AKI 1	4	4	2	NA	6	8.5-8-4		No	(a) CC/eradication (b) CC/eradication	
2	Male	(a) 24+5; 750 (b) 25+6; 840	NEC 3	*S. capitis* (1.0/0.38)	HFOV Dopa Insulin	11.5	198	AKI 1 PB	4	5	4	NA	11	8.5-12-9		A	(a) CC/eradication (b) CF/LR	
3	Female	(a) 25+4; 870 (b) 27+4; 970	NEC 2	*S. capitis* (<0.5/0.38) *S. haemolyticus* (2.0/1.5)		NR	<5	PB	7	4	7	2	7	8.5-8-8	5.9	B	(a) CC/eradication (b) CC/eradication	
4	Female	(a) 26+6; 575 (b)28+5; 754		*S. capitis* (<0.5/0.5) *Staphylococcus warnerii* (1.0/0.38)	ACV Filling	2.84	<5	AKI 2 PB	10	11	10	NA	9	3.5-8-1 8-8-8	3.6	B	(a) CC/eradication (b) CC/eradication	Cholestasis, G1, possible
5	Male	(a) 24; 600 (b) 24+5; 620	Umbilical inflammation[a]	*S. haemolyticus* (1.0/1.0)	Dopa ACV Insulin	NR	<5	AKI 1 PB	8	8	3	1	NA	6-8-7		No	(a) CF/eradication (b) NA	AKI 2, G2, possible
6	Male	(a) 24+4; 670 (b) 25; 620	VAP NEC 3	*S. haemolyticus* (1.0/1.0)	Dopa HFOV	0.84	28	VAP PB	14	14	8	NA	8	6-8-6	5.5	B	(a) CC/eradication (b) CC/eradication	
7	Female	(a) 25; 785 (b) 25+4; 720		*S. haemolyticus* (1.0/1.0)	Dopa HFOV Insulin	2.22	7.5	AKI 1 PB	9	9	8	1	NA	6-8-6 8-8-3		B	(a) CF/persistent (b) NA	
8	Male	(a) 25+1; 780 (b) 26; 795		*Staphylococcus epidermidis* (1.0/0.12) *S. warnerii* (0.25/0.12)	Insulin	2.54	10.4	PB	4	4	5	1	7	6-8-8	4.5	A	(a) CC/eradication (b) CF/ER	Diarrhea, G1, possible
9	Female	(a) 26+1; 791 (b) 27+4; 875		*S. epidermidis* (1.0/0.38)		0.58	<5	PB	3	3	4	2	6	8-8-8		B	(a) CC/eradication (b) CC/eradication	

(Continued)

TABLE 1. *(Continued.)*

Patients*	Gender	GA or PMA (Weeks) and Weight (g), respectively, (a) at Birth (b) at Sepsis	Specific Condition	*Staphylococcus* Species and Vancomycin/Ceftaroline MICs (mg/L)	Sepsis Intensity	PCT (ng/mL)	Maximal C-reactive Protein (mg/L)	Medical Reason for Ceftaroline	Duration (Days): Vancomycin	Duration (Days): Effective Vancomycin†	Duration (Days): Bacteremia Before Ceftaroline	Duration (Days): Time to Sterile Culture After CFT	Duration (Days): Antibiotics After First Negative Culture	Ceftaroline Posology (per dose in mg/kg) Interval (h) Length (d)	Ceftaroline Concentration (μg/mL)	Catheter Removal for Treatment‡	Outcome Microbiological/Clinical at (a) EOT and (b) 7 Days Later	Adverse Effects Possibly Related to Treatment, Grade (G) Severity According to CTCAE General Guidelines, Imputability According to Naranjo
10	Female	(a) 25+2; 485 (b) 28+1; 685	Severe IUGR VAP	*S. epidermidis* (2.0/0.25) *S. haemolyticus* (<0.25/1.0)	HFOV Insulin	0.82	14.8	VAP B	5	5	1/7[c]	5/2[d]	11	8-8-13		No	(a) CC/eradication (b) CC/eradication	Hypereosinophilia, G1, probable
11	Male	(a) 24; 665 (b) 26+6; 695		*S. epidermidis* (1.0/1.0) *S. haemolyticus* (2.0/1.0)	ACV Insuline	41	32	PB AKI 1	15	11	11	NA	10	8.5-8-7		B	(a) CC/eradication (b) CC/eradication	NEC (Bell 1), G2, possible
12	Female	(a) 25+6; 678 (b) 26+2; 620	Neonatal kidney injury[b]	*S. haemolyticus* (1.0/1.0)		1.93	9.3	PB AKI 3	5	3	5	NA	NA	3-8-4	3.8	A	(a) CF/persistent (b) NA	
13	Male	(a) 26+6; 900 (b) 27+3; 905	NEC 2 Severe BPD	*S. haemolyticus* (0.5/1.0)	Dopa FB ACV Insulin	NR	23.5	VAP	8	4	1	NA	11	8-8-6		B	(a) CC/eradication (b) CC/eradication	AKI 1, G1, possible Cholestasis, G3, possible
14	Female	(a) 25+6; 605 (b) 26+1; 620		*S. capitis* (1.0/0.19) *S. epidermidis* (2.0/0.25)		NR	<5	AKI2	12	10	5	NA	8	8-8-6		No	(a) CC/eradication (b) CC/eradication	AKI 2, G1, possible NEC (Bell 1), G2, possible
15	Female	(a) 25+2; 910 (b) 28+5; 715	Severe BPD VAP	*Staphylococcus aureus* (0.5/0.38)	HFOV	NR	<5	VAP	4	4	1/1[e]	NA	8	8-8-6		No	(a) CC/eradication (b) CC/eradication	Diarrhea, G1, possible Hypereosinophilia, G1, possible
16	Female	(a) 25+1; 735 (b) 28+5; 1090	*Candida bacteriema*	*S. epidermidis* (1.0/0.25)	ACV	1.07	<5	AKI 1 PB	6	6	7	3	1	8-8-4		B	(a) CC/eradication (b) CC/eradication	NEC (Bell 2), G3, possible

*Patients with clinical failure of CFT treatment are in bold font.
†Sum of days with effective vancomycinemia.
‡Catheter removal: A: after CFT treatment; B: before end of CFT.
[a]Umbilical inflammation motivating vancomycin initiation 5 days before bacteremia occurred.
[b]Kidney injury because of twin-to-twin transfusion syndrome.
[c]1 Day bacteremia and 7 days positive protected bronchial sampling.
[d]5 Days for sterile protected bronchial sampling and 2 days for sterile hemoculture.
[e]1 Day bacteremia and 1 day positive protected bronchial sampling.
ACV, assist control ventilation; AKI, acute kidney injury (stage); BPD, bronchopulmonary dysplasia; CC, clinical cure; CF, clinical failure; CTCAE, Common Terminology Criteria for Adverse Events (V5) classification; EOT, end of treatment; ER, early recurrence; FB, fluid bolus; GA, gestational age; HFOV, high frequency oscillatory ventilation; IUGR, intrauterine growth retardation; LR, late recurrence; NA, not applicable; NEC, necrotizing enterocolitis (stage); PB, persistent bacteriemia; PCT, maximal procalcitonin; PMA, postmenstrual age; VAP, ventilation-acquired pneumonia.

The limited data available suggested a dose of 6–8.5 mg/kg/8 h.[5,9] We suggest a dose of 8–8.5 mg/kg/8 h (with catheter removal if persistent bacteremia) in all patients except in the case of AKI stage 3 where we would recommend 3–4 mg/kg/8 h.[10] No clinical breakpoints and epidemiological cutoff values have been established for CoNS by EUCAST, but according to our data, these dosing regimens should successfully eradicate strains with MICs of ≤1 mg/L (EUCAST breakpoint for ceftaroline resistance in *S. aureus*).

The duration of ceftaroline treatment varies greatly. The duration of antibiotic therapy recommended in the department is 7 days from the first negative blood culture, and overall 10 days in case of pneumonia. In practice, the duration was defined according to the clinical context (change of the catheter, evolution of the child, VAP) and physician's experience. The bactericidal effect of ceftaroline and results suggest that a shorter duration is possible. Indeed, patient 16 received only 1-day ceftaroline after catheter removal and first negative blood culture. Further studies are warranted to confirm this. Also, we are eagerly awaiting the results of studies on the value of procalcitonin in defining the duration of antibiotic therapy in preterm infants.

CONCLUSIONS

To our knowledge, this is the first study reporting treatment of LOS with *Staphylococci* in preterm infants of less than 32 weeks GA and <28 days postnatal days, with a good safety and efficacy of ceftaroline. Although the number of participants is limited and the effect on microbiota should be monitored, these data provide encouraging results, and support the initiation of further studies. Moreover, because of its broad spectrum and reduced monitoring, its use could be extended to other indications.

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